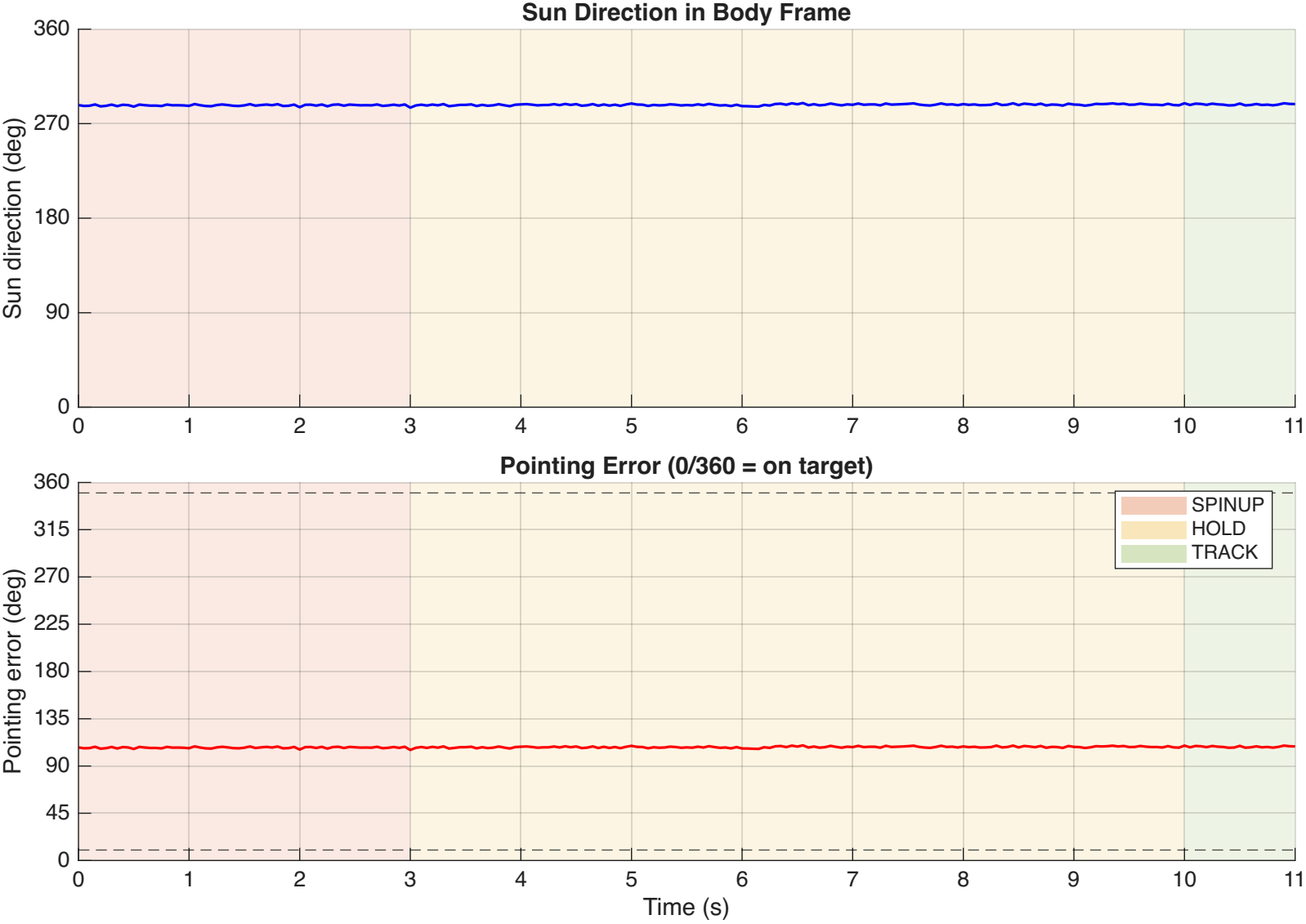
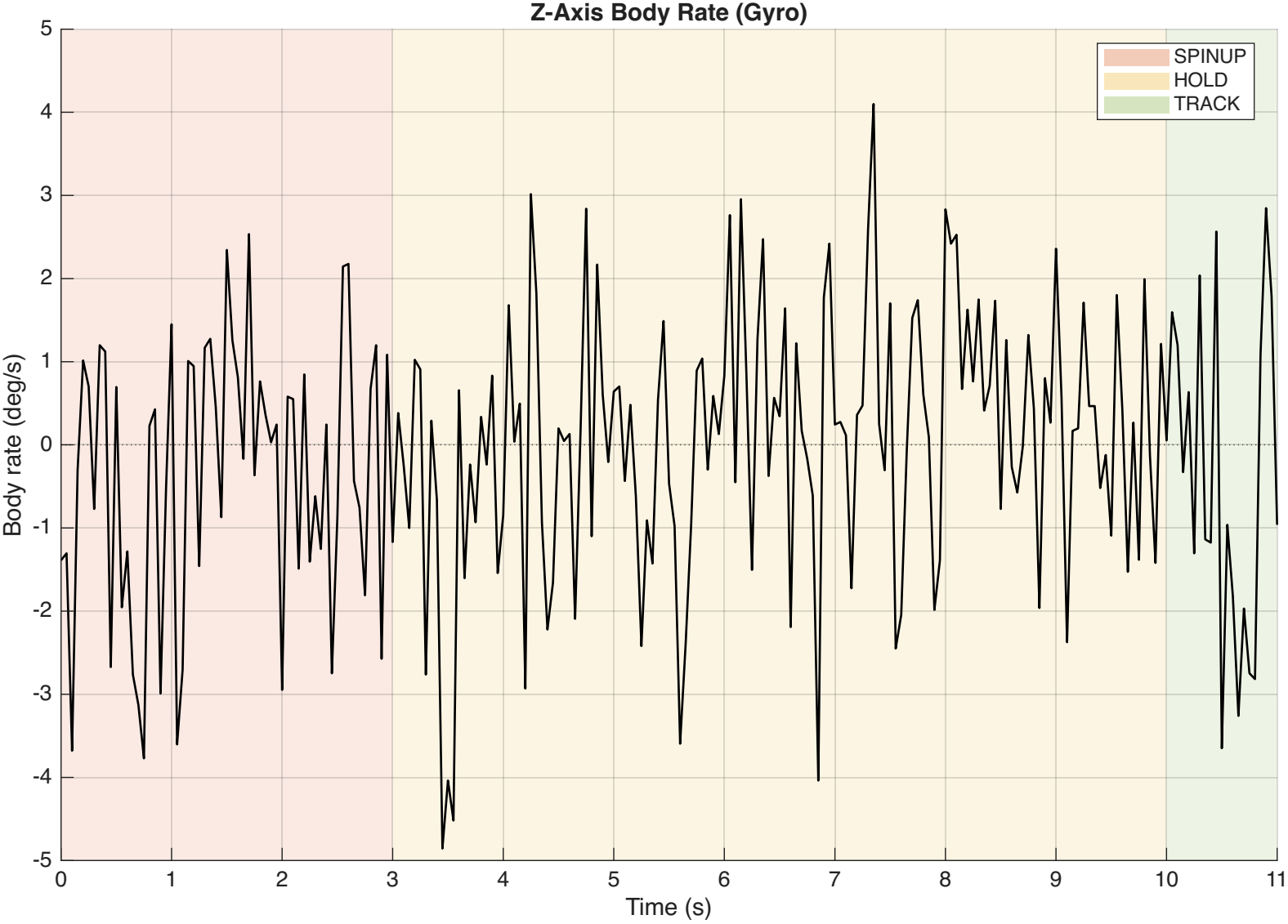
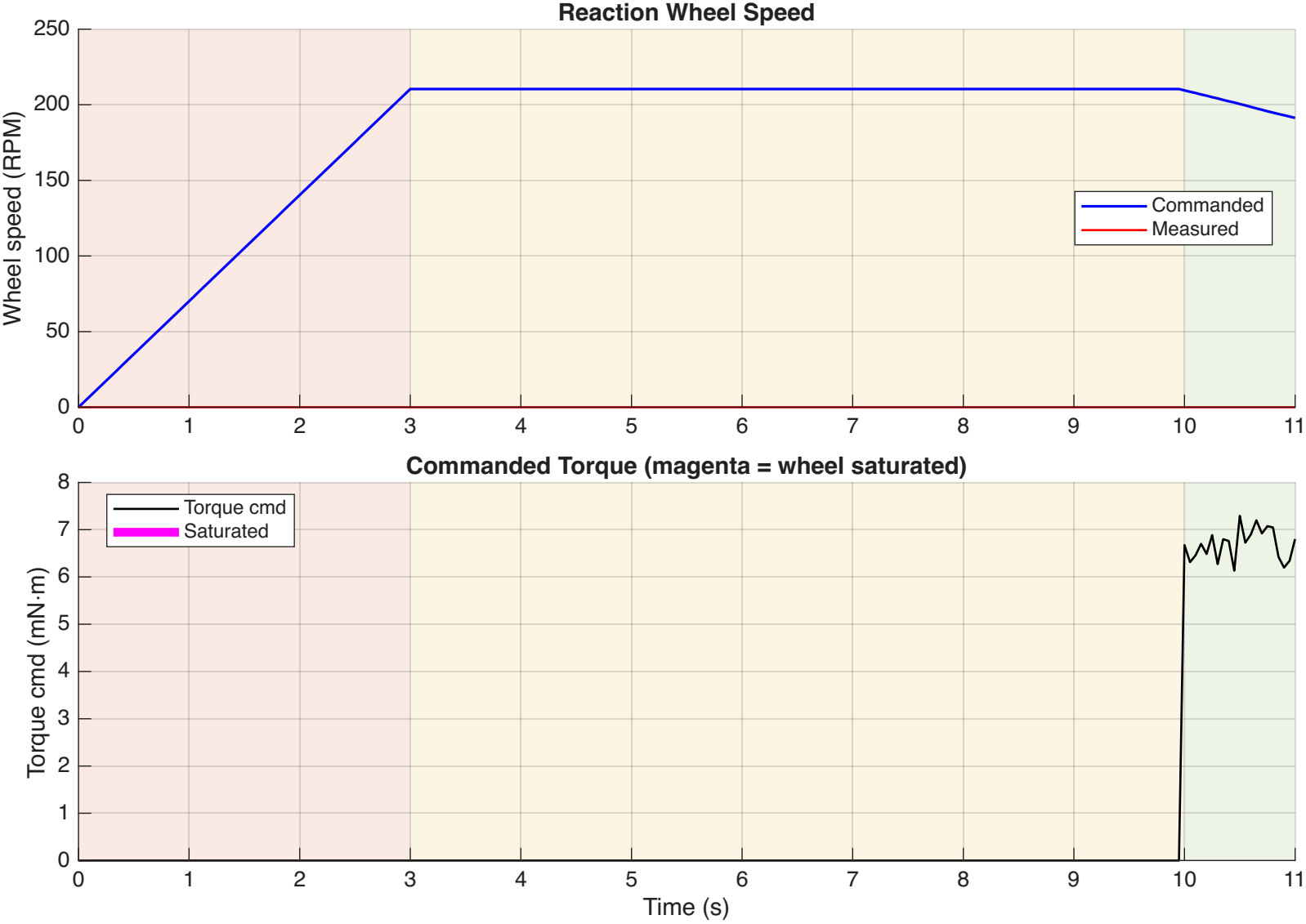
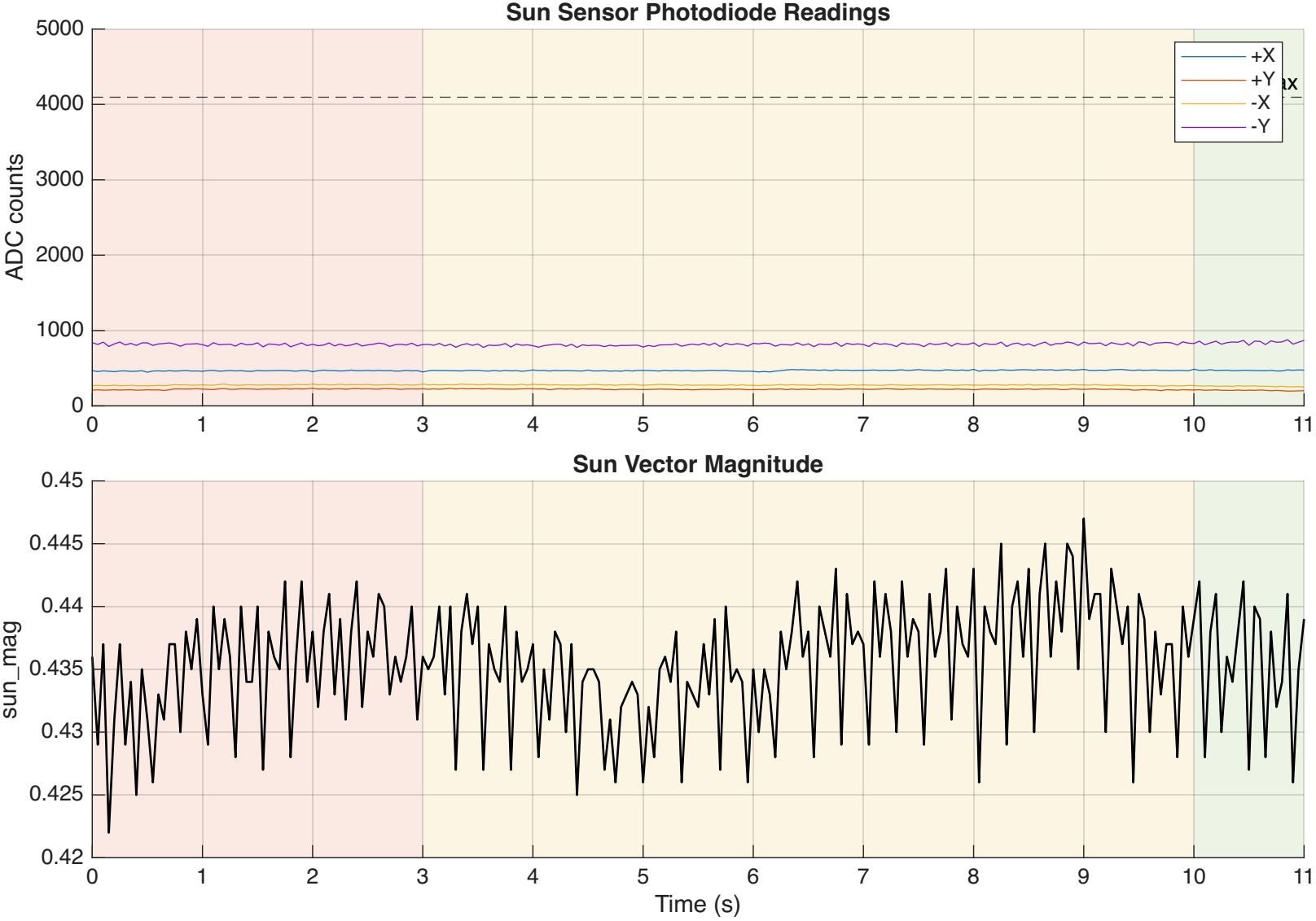
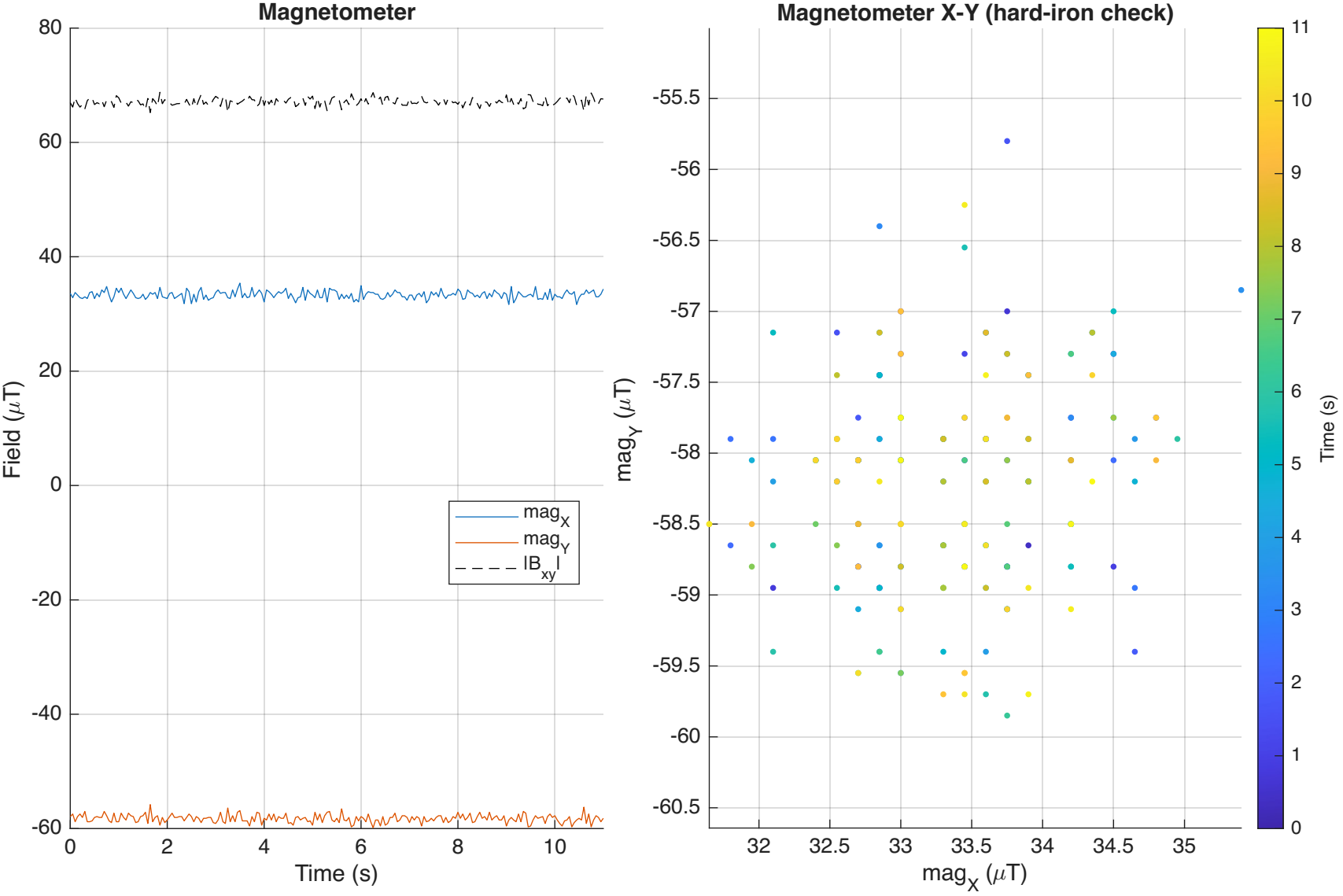
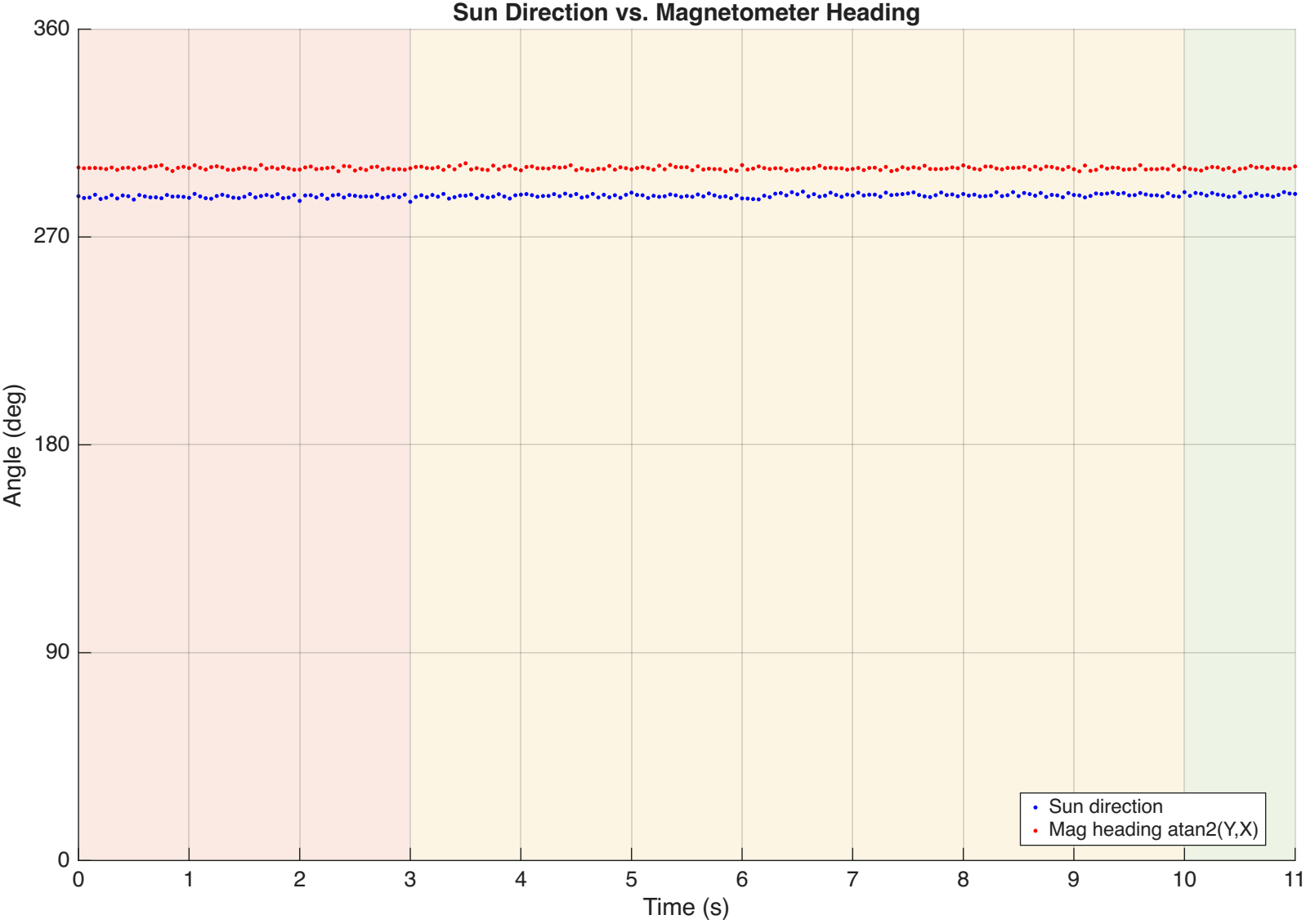

Sun Direction in Body Frame
Sun direction (deg)
360
270
180
90
0
0
1
2
3
4
5
6
7
8
9
10
11
Pointing Error (0/360 = on target)
Pointing error (deg)
360
315
270
225
180
135
90
45
0
SPINUP
HOLD
TRACK
Time (s)

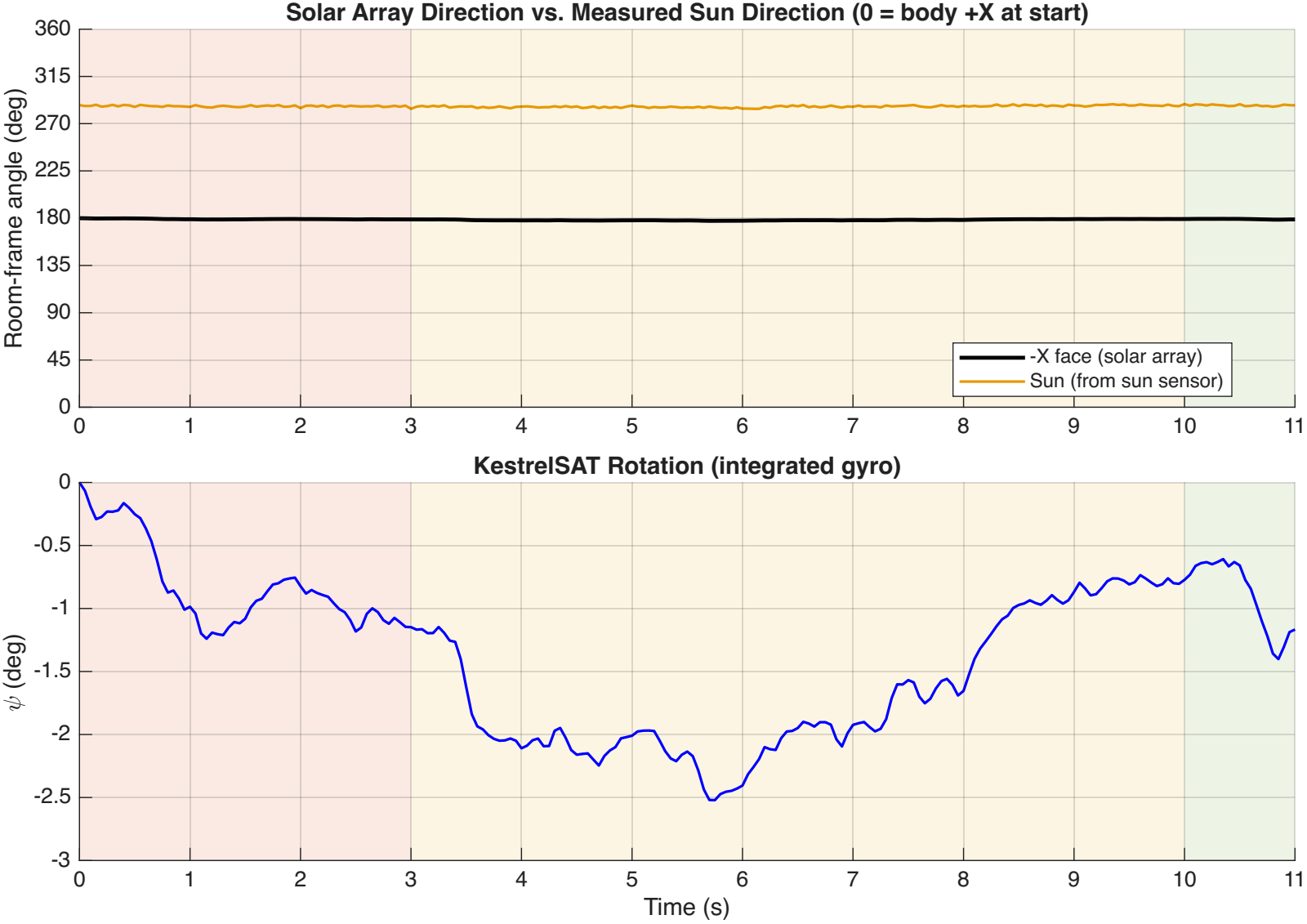

Z-Axis Body Rate (Gyro)
SPINUP
HOLD
TRACK
Body rate (deg/s)
Time (s)
5
4
3
2
1
0
-1
-2
-3
-4
-5
0
1
2
3
4
5
6
7
8
9
10
11

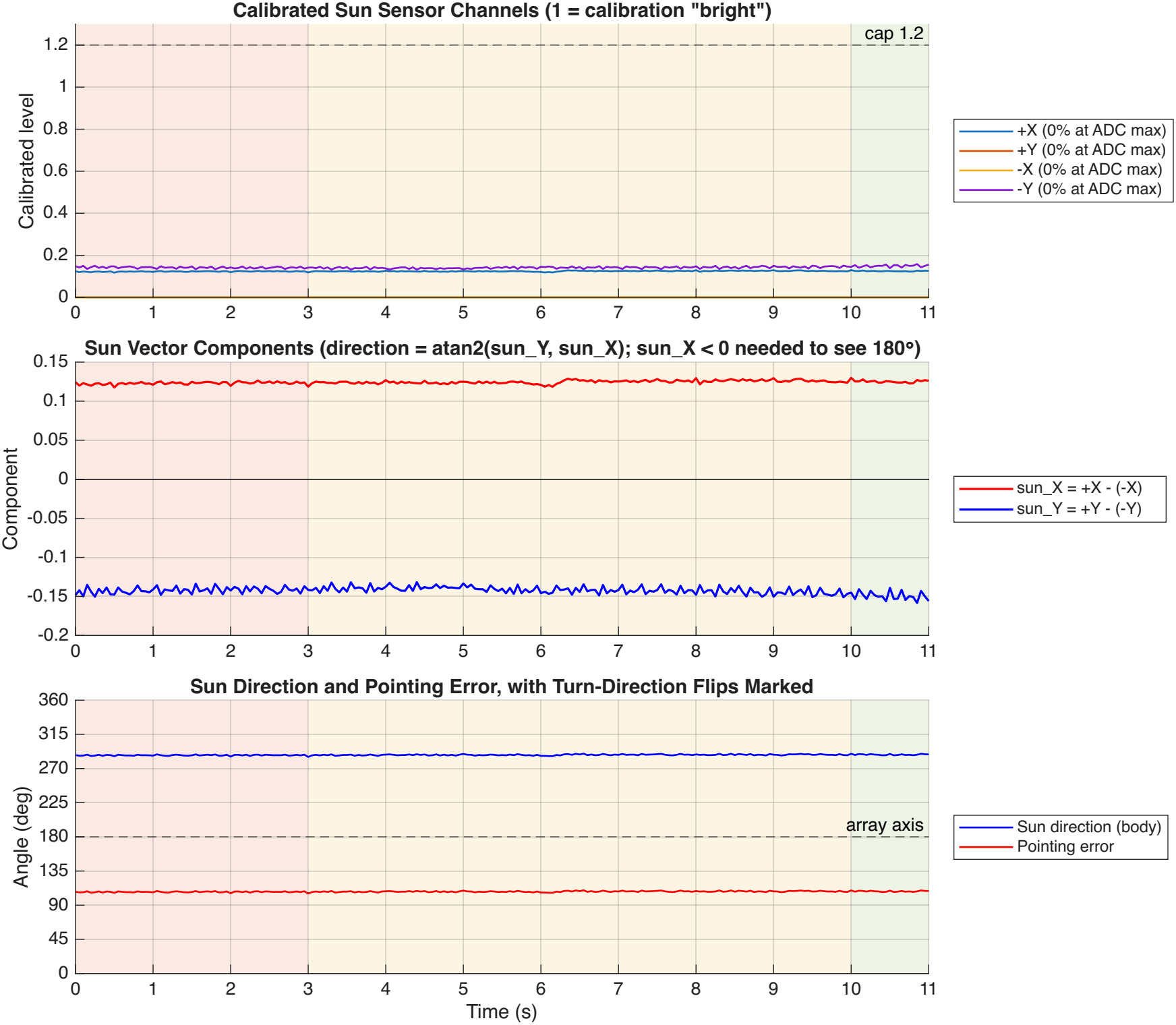

Reaction Wheel Speed
Wheel speed (RPM)
250
200
150
100
50
0
0
1
2
3
4
5
6
7
8
9
10
11
Commanded
Measured
Commanded Torque (magenta = wheel saturated)
Torque cmd (mN·m)
8
7
6
5
4
3
2
1
0
Torque cmd
Saturated
Time (s)

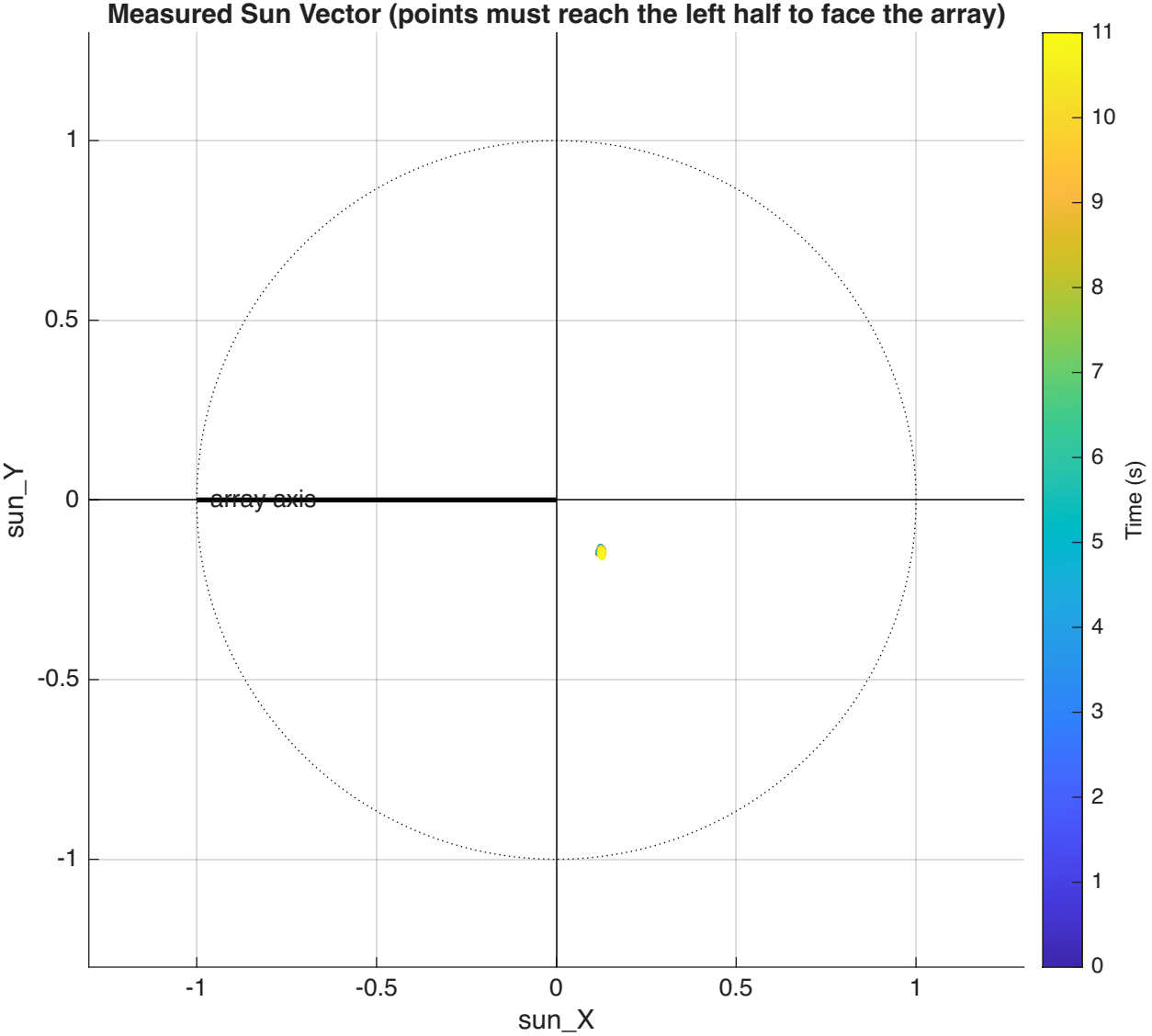

Sun Sensor Photodiode Readings
ADC counts
5000
4000
3000
2000
1000
0
+X
+Y
-X
-Y
0
1
2
3
4
5
6
7
8
9
10
11
Sun Vector Magnitude
sun_mag
0.45
0.445
0.44
0.435
0.43
0.425
0.42
Time (s)

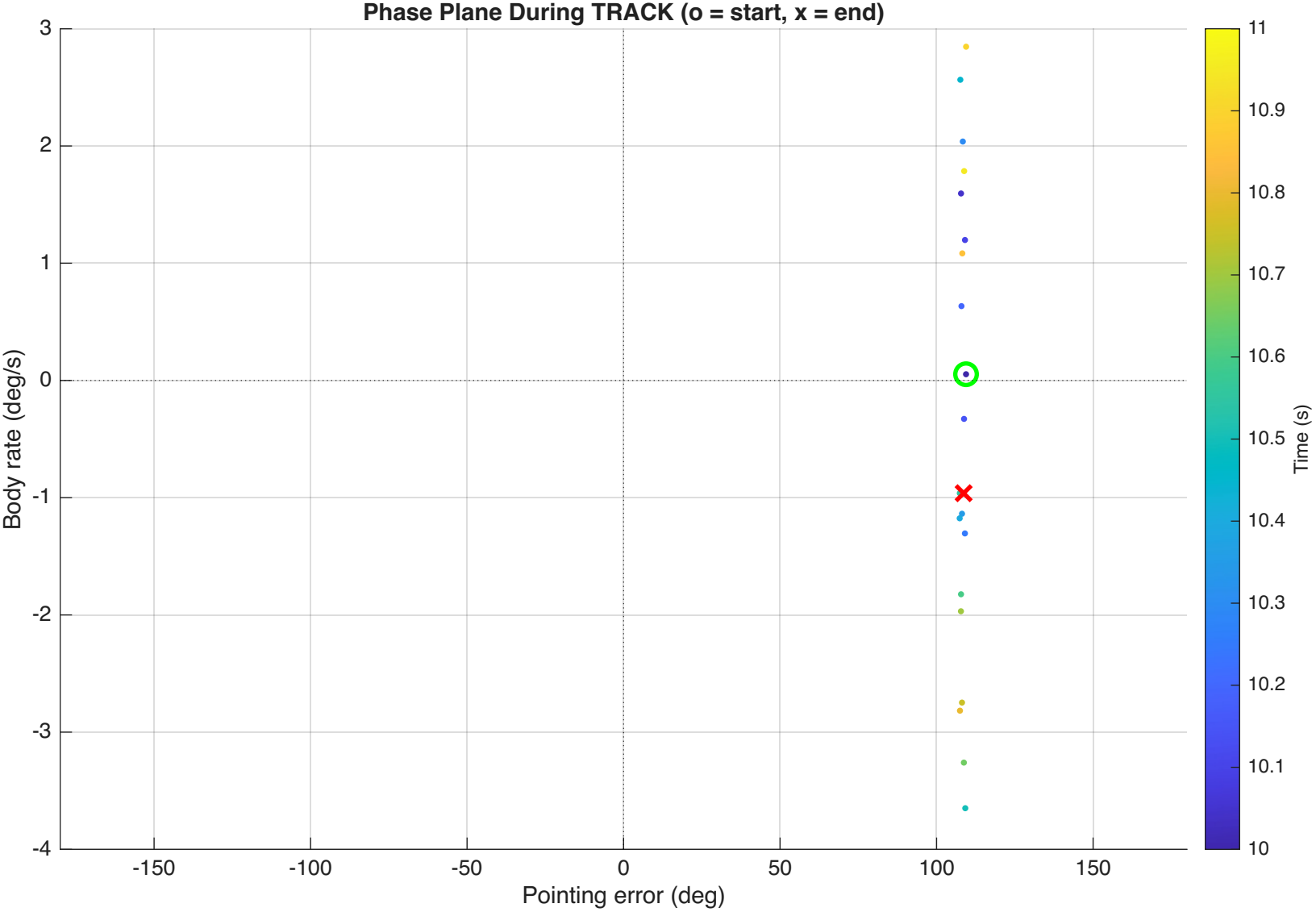

Magnetometer
Field (μT)
Time (s)
magX
magY
IBxyI
Magnetometer X-Y (hard-iron check)
magY (μT)
magX (μT)
Time (s)

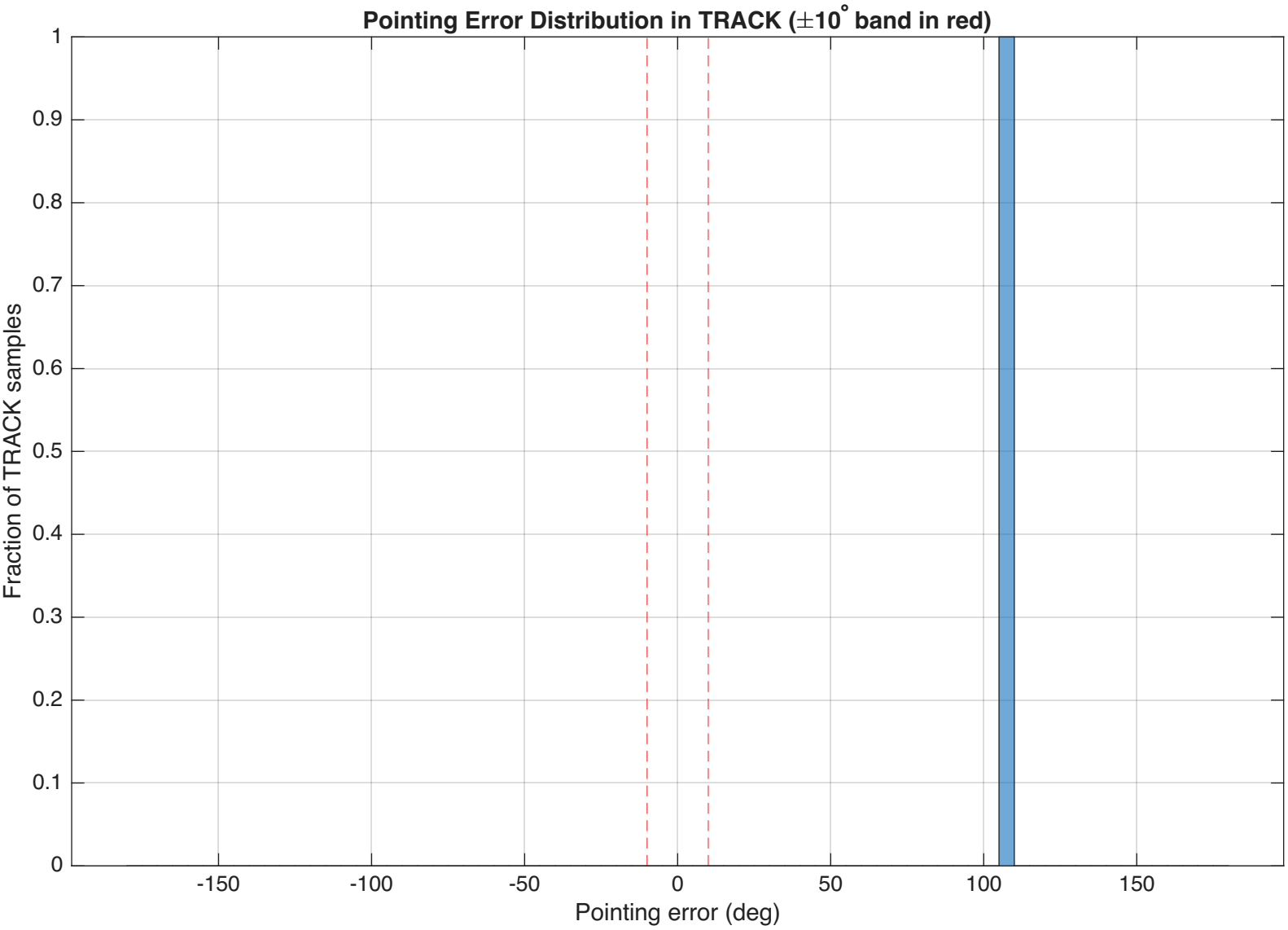

Sun Direction vs. Magnetometer Heading
360
270
180
90
0
Angle (deg)
0
1
2
3
4
5
6
7
8
9
10
11
Time (s)
Sun direction
Mag heading atan2(Y,X)


Solar Array Direction vs. Measured Sun Direction (0 = body +X at start)
Room-frame angle (deg)
360
315
270
225
180
135
90
45
0
0
1
2
3
4
5
6
7
8
9
10
11
-X face (solar array)
Sun (from sun sensor)
KestrelSAT Rotation (integrated gyro)
ψ (deg)
0
-0.5
-1
-1.5
-2
-2.5
-3
Time (s)


Calibrated Sun Sensor Channels (1 = calibration "bright")
cap 1.2
Calibrated level
+X (0% at ADC max)
+Y (0% at ADC max)
-X (0% at ADC max)
-Y (0% at ADC max)
Sun Vector Components (direction = atan2(sun_Y, sun_X); sun_X < 0 needed to see 180°)
Component
sun_X = +X - (-X)
sun_Y = +Y - (-Y)
Sun Direction and Pointing Error, with Turn-Direction Flips Marked
array axis
Angle (deg)
Sun direction (body)
Pointing error
Time (s)


Measured Sun Vector (points must reach the left half to face the array)
array axis
sun_Y
sun_X
Time (s)
1
0.5
0
-0.5
-1
-1
-0.5
0
0.5
1
11
10
9
8
7
6
5
4
3
2
1
0


Phase Plane During TRACK (o = start, x = end)
Body rate (deg/s)
Pointing error (deg)
Time (s)
3
2
1
0
-1
-2
-3
-4
-150
-100
-50
0
50
100
150
11
10.9
10.8
10.7
10.6
10.5
10.4
10.3
10.2
10.1
10


Pointing Error Distribution in TRACK (±10° band in red)
Fraction of TRACK samples
Pointing error (deg)
1
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
-150
-100
-50
0
50
100
150